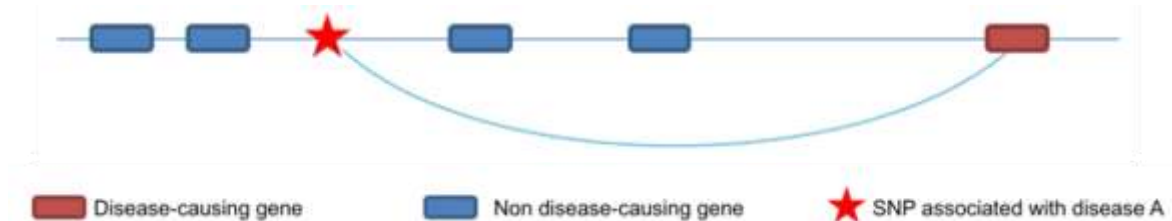


Public bioinformatics resources for GWAS interpretation

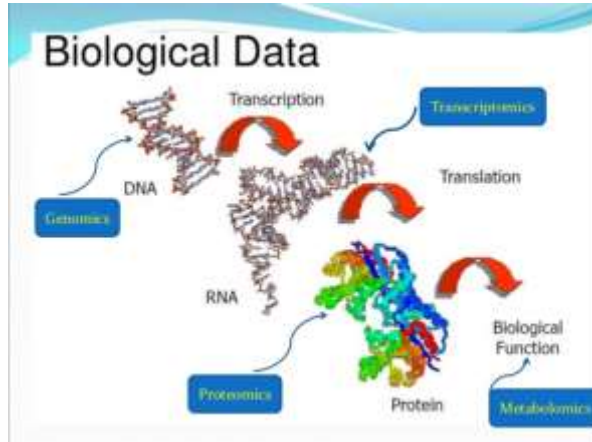
Dr. Konstantinos Hatzikotoulas
Institute of Translational Genomics

Motivation

- Greatest challenge in genetic association studies is to assign functionality (causality) to associated SNPs and genes.
- Predict the most likely genes and variants driving the phenotype associations.
- Need to integrate as much information as possible through genome annotation tools to drive the experimental validation.



What is Bioinformatics?



Computers aid to collect, intergrade and analyse biological data ->
Optimal biological interpretation



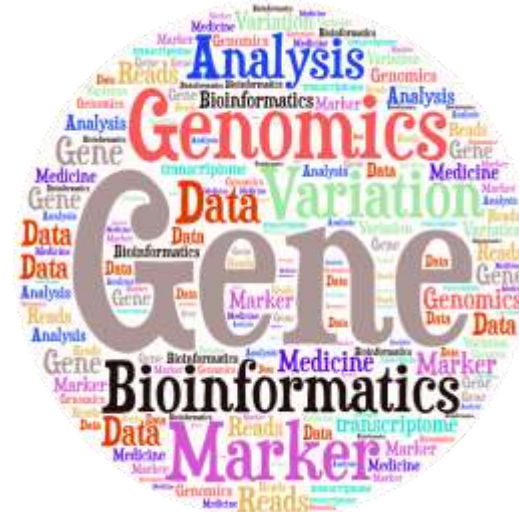
Goal of Bioinformatics

- Develop **databases** and **computational tools** to generate knowledge to better understanding a living cell and how it functions at molecular level.



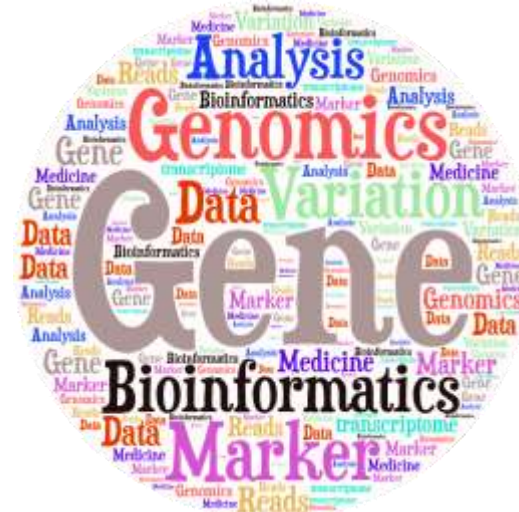
Publicly available bioinformatics resources

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



Publicly available bioinformatics resources

- **Genome Browsers**
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



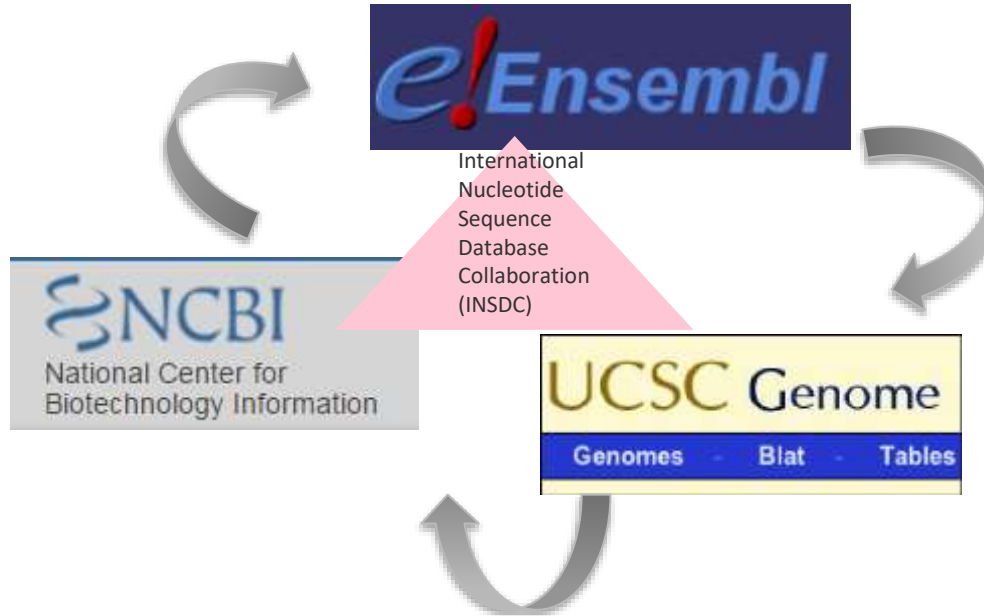
Genome Browsers

- Ensembl : <http://www.ensembl.org>
- UCSC: <http://genome.ucsc.edu/>
- NCBI: <https://www.ncbi.nlm.nih.gov/>



Genome Browsers

- Agreement: To begin bringing all genome assembly data into compliance moving forward



Ensembl ... more than a browser!

- Ensembl is a joint project between the EBI (European Bioinformatics Institute) and the Wellcome Trust Sanger Institute
- **What can I do with Ensembl?**
 - Examine single nucleotide polymorphisms (SNPs) for a gene or chromosomal region.
 - View SNPs across strains (rat, mouse), populations (human) or breeds (dog).
 - View genes with annotations along the chromosome.
 - View alternative transcripts (i.e. splice variants) for a given gene.
 - Upload your own data.
 - Export sequence or create a table of gene information with BioMart.
 - Determine how your variants affect genes and transcripts using the Variant Effect Predictor.
 - Share Ensembl views with your colleagues and collaborators.



Ensembl ... more than a browser!

- Ensembl is a joint project between the EBI (European Bioinformatics Institute) and the Wellcome Trust Sanger Institute
- **What can I do with Ensembl?**
 - Examine single nucleotide polymorphisms (SNPs) for a gene or chromosomal region.
 - View SNPs across strains (rat, mouse), populations (human) or breeds (dog).
 - View genes with other annotation along the chromosome.
 - View alternative transcripts (i.e. splice variants) for a given gene.
 - Upload your own data.
 - Export sequence or create a table of gene information with BioMart.
 - Determine how your variants affect genes and transcripts using the Variant Effect Predictor.
 - Share Ensembl views with your colleagues and collaborators.



Ensembl ... the front page

- It contains lots of information and links to help you navigate Ensembl.

The screenshot shows the Ensembl homepage with several key features highlighted by callouts:

- Link back to homepage:** Points to the Ensembl logo in the top navigation bar.
- Ensembl tools:** Points to the 'Tools' link in the top navigation bar.
- Blue bar remains visible on every page:** Points to the dark blue header bar containing navigation links.
- Search:** Points to the search bar in the top right corner.
- See the current release number and what's new:** Points to the 'Ensembl Release 108 (Apr 2022)' section on the right.
- Annotations:** A red arrow points to the 'Favorite genomes' section, and a blue arrow points to the 'Pig breeds' section.

The main content area includes a search bar, a list of genomes (Pig breeds, Mouse, Zebrafish), and a section for 'Ensembl Rapid Release' with a list of new assemblies.

Hint: <https://www.ensembl.org/index.html>

Ensembl ... the front page ... Exercise 1

1. Go to the species homepage for **Giant Panda**. What is the name of the genome assembly for Panda?
2. How long is the Panda genome (in bp)? How many coding genes have been annotated?
3. Repeat for **Human**.



- Let's have a look at a specific variant, rs143383 (chr20:35438203).

[BLAST/BLAT](#) | [VEP](#) | [Tools](#) | [BioMart](#) | [Downloads](#) | [Help & Docs](#) | [Blog](#)

Human (GRCh38.p13)

Location: 20,35,437,703-35,438,703
Variant: rs143383
Jobs

Variant displays
Explore this variant
Genomic context
Genes and regulation
Flanking sequence
Population genetics
Phenotype data
Sample genotypes
Linkage disequilibrium
Phylogenetic context
Citations
3D Protein model
Configure this page
Custom tracks
Export data
Share this page
Bookmark this page

rs143383 SNP

Most severe consequence: 5 prime UTR variant | [See all predicted consequences](#)

Alleles: G/A | Ancestral: G | Highest population MAF: 0.49

Change tolerance: CADD: A:21.4 | GERP: 2.76

Location: [Chromosome 20:35438203](#) (forward strand) | VCF: [20:35438203](#)

Co-located variants: dbSNP [rs185582358](#) (G/A) | HGMD-PUBLIC [CMR72309](#)

Evidence status:

Clinical significance:

HGVs names: This variant has 5 HGVs names - [Show](#)

Synonyms: This variant has 8 synonyms - [Show](#)

Genotyping chips: This variant has assays on 5 chips - [Show](#)

Original source: Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#)

About this variant: This variant overlaps 2 transcripts, has 3009 sample genotypes, is associated with 11 phenotypes and is mentioned in 201 citations.

Description from SNPedia: This SNP is associated with [cellophaneitis# \(DAL\)](#). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding [the embryonic stem-specific GDF5](#) (also known as [cartilage-derived embryonic development 5](#) or [CDER5](#)) ([MIM:608042](#)) - [Show](#)

Genomic context
 Genes and regulation
 Flanking sequence
 Population genetics
 Phenotype data
 Sample genotypes
 Linkage disequilibrium
 Phylogenetic context
 Citations
 3D Protein model

Variant information

Variation icons. These go to the same places as the links on the left

Ensembl ... Exploring variants ...

Variant types

➤ What type of variants is rs143383 (chr20:35438203)?

Ensembl | BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▾

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▾

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Configure this page Custom tracks Expert data Share this page Bookmark this page

rs143383 SNP

Most severe consequence: 5 prime UTR variant | See all predicted consequences

Alleles: G/A | Ancestral: G | Highest population MAF: 0.49

Change tolerance: CADD: A:21.4 | GERP: 2.76

Location: Chromosome 20:35438203 (forward strand) | VCF: 20:35438203:G:A

Co-located variants: dbSNP rs195582358 (G/A) | HGMD-PUBLIC CB072209

Evidence status: ClinVar, dbSNP, Ensembl, RefSeq, UniProt, AD

Clinical significance: This variant has 5 HGVS names - Show

HGVs names: This variant has 9 synonyms - Show

Synonyms: This variant has assays on 5 chips - Show

Genotyping chips: Variants (including SNPs and indels) imported from dbSNP (release 154) | View in dbSNP

Original source: This variant overlaps 2 transcripts, has 3009 sample genotypes, is associated with 11 phenotypes and is mentioned in 201 citations.

About this variant: This SNP is associated with osteoarthritis (OA). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding the embryonic stage oncofetus. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... Show

Description from SNPedia

Explore this variant

- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Ensembl ... Exploring variants ...

Variant types

1. Small scale in one or few nucleotides of a gene

Type	Description	Example (Reference / Alternative)
SNP	Single Nucleotide Polymorphism	Ref. ...TTG A CGTA... Alt. ...TTG G CGTA...
Insertion	Insertion of one or several nucleotides	Ref. ...TTGACGTA... Alt. ...TTGAT GCG CGTA...
Deletion	Deletion of one or several nucleotides	Ref. ...TTG AC GTA... Alt. ...TTGGTA...
Indel	An insertion and a deletion, affecting 2 or more nucleotides	Ref. ...TTG AC GTA... Alt. ...TTG GCT CGTA...
Substitution	A sequence alteration where the length of the change in the variant is the same as that of the reference.	Ref. ...TTG AC GTA... Alt. ...TTG TAG GTA...

2. Large scale in chromosomal structure

CNV

Copy Number Variation: increases or decreases the copy number of a given region

Reference:



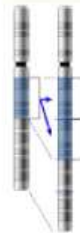
"Gain" of one copy:



"Loss" of one copy:



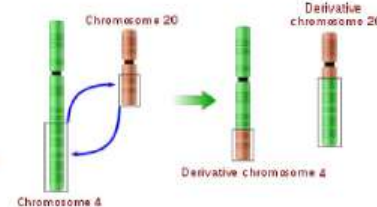
deletion



duplication



insertion

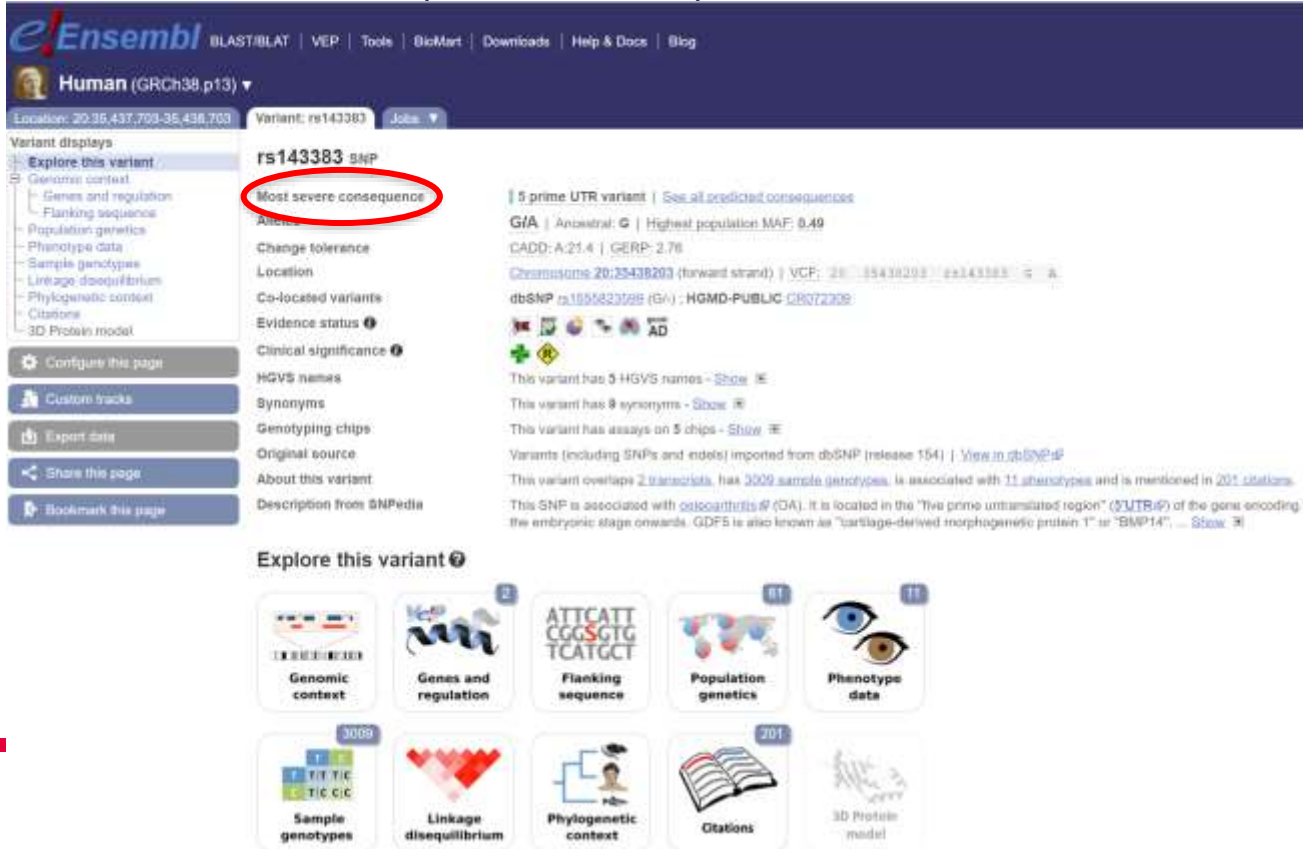


translocation

Ensembl ... Exploring variants ...

Calculated variant consequences

➤ Let's have a look at the predicted consequences of the variant.



Ensembl | BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▾

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▾

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Configure this page Custom tracks Expert data Share this page Bookmark this page

rs143383 SNP

Most severe consequence

5 prime UTR variant | See all predicted consequences

G1A | Ancestral: G | Highest population MAF: 0.49

CADD: A:21.4 | GERP: 2.76

Chromosome 20:35438203 (forward strand) | VCF: 20:35438203:G>A

dbSNP rs195582358 (G/A) | HGMD-PUBLIC CB072909

AD

This variant has 5 HGVS names - [Show](#)

This variant has 9 synonyms - [Show](#)

This variant has assays on 5 chips - [Show](#)

Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#)

This variant overlaps 2 transcripts, has 3009 sample genotypes, is associated with 11 phenotypes and is mentioned in 201 citations.

This SNP is associated with [colicathritis](#) (GIA). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding the embryonic stage oncof. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... [Show](#)

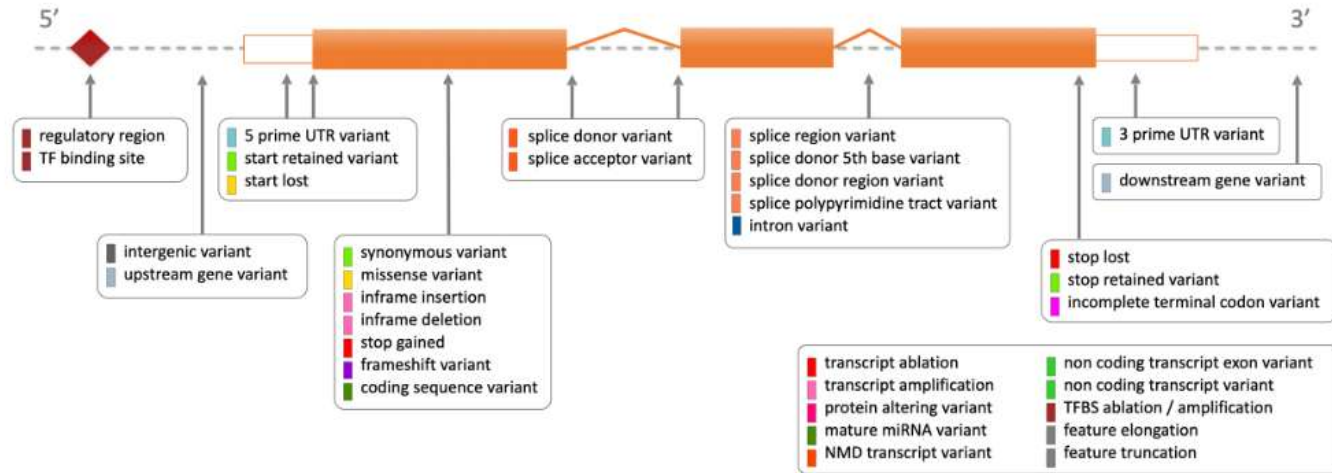
Explore this variant

- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Ensembl ... Exploring variants ...

Calculated variant consequences

- A rule-based approach to predict the effects that each allele of the variant may have on the transcript.



http://www.ensembl.org/info/genome/variation/prediction/predicted_data.html

HELMHOLTZ RESEARCH FOR GRAND CHALLENGES

Ensembl ... Exploring variants ...

Calculated variant consequences

➤ Let's have a look at all the predicted consequences of the variant.

The screenshot displays the Ensembl genome browser interface for variant rs143383. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. The main header shows the Human (GRCh38.p13) genome and the variant location: 20:35,437,703-35,438,703. The variant is identified as rs143383 SNP.

On the left, the 'Variant displays' sidebar lists various tracks: Genomic context, Genes and regulation, Flanking sequence, Population genetics, Phenotype data, Sample genotypes, Linkage disequilibrium, Phylogenetic context, Citations, and 3D Protein model. Below these are options to 'Configure this page', 'Custom tracks', 'Export data', 'Share this page', and 'Bookmark this page'.

The main content area shows the variant details for rs143383. It lists the most severe consequence as '5 prime UTR variant' and provides a link to 'See all predicted consequences', which is circled in red. Other details include the variant's location on chromosome 20, its VCF representation, and its association with dbSNP and HGMD-PUBLIC. The variant is also associated with 5 HGVS names, 9 synonyms, and 5 assays.

Below the variant details, the 'Explore this variant' section features a grid of 11 tracks, each with an icon and a count: Genomic context (2), Genes and regulation (2), Flanking sequence (2), Population genetics (11), Phenotype data (11), Sample genotypes (3009), Linkage disequilibrium (3009), Phylogenetic context (201), Citations (201), and 3D Protein model (201). The 'Genes and regulation' track is circled in red.



Ensembl ... Exploring variants ...

Calculated variant consequences

➤ This variant is found in 2 transcripts of the *GDF5* gene.

Genes and regulation

Gene and Transcript consequences



Gene	Transcript (abnvt)	Allele (Tr. allele)	Consequence Type	Position in transcript	Position in CDS	Position in protein	AA	Codons	Detail
ENSG00000125963	ENSG00000125963.1	A	5 prime UTR variant	26 out of 2005					RefSeq
HMSC-0078	HMSC-0078.1	A	5 prime UTR variant						RefSeq



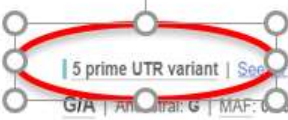
coding_sequence_variant	A sequence variant that changes the coding sequence	SO:0001380.6	Coding sequence variant	MODIFIER
mature_miRNA_variant	A transcript variant located with the sequence of the mature miRNA	SO:0001600.6	Mature miRNA variant	MODIFIER
5_prime_UTR_variant	A UTR variant of the 5' UTR	SO:0001620.6	5 prime UTR variant	MODIFIER
3_prime_UTR_variant	A UTR variant of the 3' UTR	SO:0001621.6	3 prime UTR variant	MODIFIER
non_coding_transcript_exon_variant	A sequence variant that changes non-coding exon sequence in a non-coding transcript	SO:0001792.6	Non coding transcript exon variant	MODIFIER
intron_variant	A transcript variant occurring within an intron	SO:0001822.6	Intron variant	MODIFIER
NMD_transcript_variant	A variant in a transcript that is the target of NMD	SO:0001821.6	NMD transcript variant	MODIFIER
non_coding_transcript_variant	A transcript variant of a non coding RNA gene	SO:0001819.6	Non coding transcript variant	MODIFIER



Variant: rs143383 Jobs

rs143383 SNP

Most severe consequence
Alleles
Change tolerance



5 prime UTR variant | See predicted consequences

G/A | Ancestral: G | MAF: 0.00 (A) | Highest population MAF: 0.49

CADD: A:20.6 | GERP: 2.34

Ensembl ... Exploring variants...

Population genetics

➤ Let's have a look at population genetics.

The screenshot displays the Ensembl genome browser interface for the variant rs143383. The left sidebar shows the 'Variant displays' menu with 'Population genetics' highlighted. The main content area shows the variant details, including its location on chromosome 20, its frequency in the population (MAF: 0.49), and its association with various diseases (e.g., AD). The 'Explore this variant' section at the bottom features several icons representing different data types, with 'Population genetics' highlighted by a red circle.

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Variant: rs143383 SNP

Most severe consequence: 5 prime UTR variant | See all predicted consequences

Alleles: G/A | Ancestral: G | Highest population MAF: 0.49

Change tolerance: CADD: A:21.4 | GERP: 2.76

Location: Chromosome 20:25438203 (forward strand) | VCF: 20:25438203:G>A

Co-located variants: SNP rs19558231 | HGMD-PUBLIC CB072009

Evidence status: 1

Clinical significance: 1

HGVs names: This variant has 5 HGVs names - Show

Synonyms: This variant has 9 synonyms - Show

Genotyping chips: This variant has assays on 5 chips - Show

Original source: Variants (including SNPs and indels) imported from dbSNP (release 154) | View in dbSNP

About this variant: This variant overlaps 2 transcripts, has 3009 sample genotypes, is associated with 11 phenotypes and is mentioned in 201 citations. This SNP is associated with osteoarthritis (OA). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding the embryonic stage onwards. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... Show

Explore this variant

- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Ensembl ... Exploring variants ...

Population genetics

- Frequencies are grouped by project for studies with multiple subpopulations.
- Pie charts can be displayed only for 1000 Genomes allele frequencies.

Population genetics

1000 Genomes Project Phase 3 allele frequencies



AFR sub-populations



EAS sub-populations



EUR sub-populations



SAS sub-populations



Tables of more detailed data

EAS	G	0.287 (100)	A	0.713 (100)	G	0.005 (4)	A	0.995 (96)	A	0.000 (0)
EUR	G	0.344 (54)	A	0.656 (100)	G	0.148 (15)	A	0.852 (95)	A	0.400 (38)
CHB	G	0.235 (4)	A	0.765 (100)	G	0.038 (8)	A	0.962 (92)	A	0.300 (30)
CHS	G	0.216 (8)	A	0.784 (100)	G	0.047 (5)	A	0.953 (95)	A	0.110 (11)
JPT	G	0.231 (4)	A	0.769 (100)	G	0.037 (7)	A	0.963 (93)	A	0.130 (13)
KHV	G	0.308 (2)	A	0.692 (100)	G	0.011 (1)	A	0.989 (99)	A	0.140 (5)

gnomAD genomes v2.1.2 (11)

Population	Allele frequency (count)
gnomADg ALL	G: 0.522 (79301) A: 0.477 (72460)
gnomADg afr	G: 0.879 (36292) A: 0.121 (5005)
gnomADg amr	G: 0.401 (365) A: 0.599 (545)
gnomADg eur	G: 0.376 (5726) A: 0.624 (9491)
gnomADg eusj	G: 0.455 (1574) A: 0.545 (1890)
gnomADg eos	G: 0.283 (1452) A: 0.717 (3710)
gnomADg fin	G: 0.415 (4371) A: 0.585 (6163)
gnomADg gnd	G: 0.494 (116) A: 0.506 (116)
gnomADg nla	G: 0.377 (25646) A: 0.623 (42308)
gnomADg oth	G: 0.486 (1018) A: 0.514 (1072)
gnomADg sas	G: 0.559 (2733) A: 0.441 (2128)

Jump to: 1000 Genomes Project Phase 3 (132) | gnomAD genomes v2.1.2 (11) | HCB ALFA (12) | TOPMed (1) | Gambia Genome Variation Project (5)

Ensembl ... Exploring variants...

Phenotype/Disease data

➤ Let's have a look at the phenotypes associated with this variant.

The screenshot displays the Ensembl genome browser interface for variant rs143383. The left sidebar shows navigation options, with 'Phenotype data' highlighted. The main content area provides detailed information about the variant, including its location on chromosome 20, its classification as a 5' prime UTR variant, and its association with the GDF5 gene. The 'Explore this variant' section at the bottom offers a grid of links to various genomic data tracks, with 'Phenotype data' circled in red.

Ensembl | BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▾

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▾

Variant displays

- Explore this variant ▾
 - Genomic context
 - Genes and regulation
 - Flanking sequence
 - Phylogenetic context
 - Phenotype data
 - Linkage disequilibrium
 - Phylogenetic context
 - Citations
 - 3D Protein model
- Configure this page
- Custom tracks
- Expert data
- Share this page
- Bookmark this page

rs143383 SNP

Most severe consequence: 5' prime UTR variant | See all predicted consequences

Alleles: G/A | Ancestral: G | Highest population MAF: 0.49

Change tolerance: CADD: A:21.4 | GERP: 2.76

Location: Chromosome 20:35438203 (forward strand) | VCF: 20...35438203...G A

Co-located variants: dbSNP rs185412358 (G/A) | HGMD-PUBLIC CB072909

Evidence status:

Clinical significance:

HGVs names: This variant has 5 HGVs names - [Show](#)

Synonyms: This variant has 9 synonyms - [Show](#)

Genotyping chips: This variant has assays on 5 chips - [Show](#)

Original source: Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#)

About this variant: This variant overlaps 2 transcripts, has 3009 sample genotypes, is associated with 11 phenotypes and is mentioned in 201 citations.

Description from SNPedia: This SNP is associated with osteoarthritis (OA). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding the embryonic stage oncofetus. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... [Show](#)

Explore this variant

- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Ensembl ... Exploring variants...

Phenotype/Disease data

➤ This variant is associated with 15 phenotypes/diseases.

Phenotype Data ⓘ

Significant association(s)

Phenotype, disease and trait	Source(s)	Mapped Terms	Ontology Accessions	Supporting evidence	External reference	Clinical significance	Reported gene(s)	Associated allele	Statistics
ACROMEGALIC DYSPLASIA 2B	ClinVar [Humana Laboratory ...]	-	Orphanet:534374	-	-	★☆☆☆☆	LOC109461476 GDF3	-	-
Acromegalic dysplasia 2C, Humera-Thompson type	ClinVar [Humana Laboratory ...]	-	-	-	-	★☆☆☆☆	LOC109461476 GDF3	-	-
Brachydactyly	ClinVar [Humana Laboratory ...]	Brachydactyly	HP:0011564 Orphanet:2949224	-	-	★☆☆☆☆	LOC109461476 GDF3	-	-
ClinVar: phenotype not specified	ClinVar [GeneDx/Invitae]	-	-	-	-	★☆☆☆☆	LOC109461476 GDF3	-	-
Cortical surface area	NHGRI-EBI GWAS catalog	-	EFO:00107364	-	PMID:34562734	-	-	G	p-value: 1.00e-17
Goda syndrome	ClinVar [Humana Laboratory ...]	-	Orphanet:20984	-	-	★☆☆☆☆	LOC109461476 GDF3	-	-
Height	GENET	body height	EFO:00043394	-	-	-	-	G	p-value: 2.81e-35
Height	NHGRI-EBI GWAS catalog	body height	EFO:00043394	-	PMID:34270784	-	GDF3	-	p-value: 1.00e-8 beta coefficient: 0.48 unit increase
Height	NHGRI-EBI GWAS catalog	body height	EFO:00043394	-	PMID:315623484	-	GDF3	-	p-value: 5.00e-78 beta coefficient: 0.06367727 unit increase
Knee osteoarthritis	NHGRI-EBI GWAS catalog	Knee osteoarthritis	EFO:00046184 HP:00552654	-	PMID:309647484	-	GDF3	-	p-value: 5.00e-7 odds ratio: 1.0495905
Knee osteoarthritis	NHGRI-EBI GWAS catalog	Knee osteoarthritis	EFO:00046184 HP:00552654	-	PMID:303740684	-	-	I	p-value: 8.00e-18 odds ratio: 1.09287
MULTIPLE SYNDYOSTOSES SYNDROME 2	ClinVar [Humana Laboratory ...]	-	Orphanet:2224	-	-	★☆☆☆☆	LOC109461476 GDF3	-	-
Osteoarthritis of hip	ClinVar [OMIM]	-	-	PMID:173846414	MM:6211464	★☆☆☆☆	LOC109461476 GDF3	-	-
Vertex-wise cortical surface area	NHGRI-EBI GWAS catalog	-	EFO:00107364	-	PMID:348135254	-	-	G	p-value: 2.00e-16 beta coefficient: 8.25 z score increase
Whole brain free water diffusion multivariate analysis	NHGRI-EBI GWAS catalog	brain measurement	EFO:00046184	-	PMID:355059524	-	-	-	p-value: 1.20e-15

Ensembl ... Exploring variants...

Citations

➤ Let's have a look at the citations of the variant.

The screenshot displays the Ensembl genome browser interface for variant rs143383. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. The main header shows the Human (GRCh38.p13) genome with the variant location 20:35,437,703-35,438,703 and the variant ID rs143383. The left sidebar, titled 'Variant displays', lists various tracks: Genomic context, Genes and regulation, Flanking sequence, Population genetics, Phenotype data, Sample genotypes, Linkage disequilibrium, Phylogenetic context, and Citations. The 'Citations' link is circled in red. Below the sidebar are buttons for 'Configure this page', 'Custom tracks', 'Export data', 'Share this page', and 'Bookmark this page'. The main content area shows the variant details for rs143383, including its most severe consequence (5 prime UTR variant), alleles (G/A), change tolerance (CADD: A:21.4 | GERP: 2.76), location (Chromosome 20:35438203 (forward strand) | VCF: 20...35438203...G A), co-located variants (dbSNP: rs195582358 (G/A) | HGMD-PUBLIC CB07209), evidence status (ClinVar, dbSNP, ClinGen, AD), clinical significance (ClinVar, dbSNP, ClinGen, AD), HGVS names, synonyms, genotyping chips, original source, and about this variant. The 'Explore this variant' section at the bottom features a grid of tiles for Genomic context, Genes and regulation, Flanking sequence, Population genetics, Phenotype data, Sample genotypes, Linkage disequilibrium, Phylogenetic context, Citations, and 3D Protein model. The 'Citations' tile is circled in red.

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▼

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▼

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations

Configure this page Custom tracks Export data Share this page Bookmark this page

rs143383 SNP

Most severe consequence: 5 prime UTR variant | See all predicted consequences

Alleles: G/A | Ancestral: G | Highest population MAF: 0.49

Change tolerance: CADD: A:21.4 | GERP: 2.76

Location: Chromosome 20:35438203 (forward strand) | VCF: 20...35438203...G A

Co-located variants: dbSNP: rs195582358 (G/A) | HGMD-PUBLIC CB07209

Evidence status: ClinVar, dbSNP, ClinGen, AD

Clinical significance: ClinVar, dbSNP, ClinGen, AD

HGVS names: This variant has 5 HGVS names - Show

Synonyms: This variant has 9 synonyms - Show

Genotyping chips: This variant has assays on 5 chips - Show

Original source: Variants (including SNPs and indels) imported from dbSNP (release 154) | View in dbSNP

About this variant: This variant overlaps 2 transcripts, has 3009 sample genotypes, is associated with 11 phenotypes and is mentioned in 201 citations. This SNP is associated with osteoarthritis (OA). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding the embryonic stage oncofetus. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... Show

Explore this variant

- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Ensembl ... Exploring variants ... Citations

- This variant is mentioned in 232 citations.

Citations

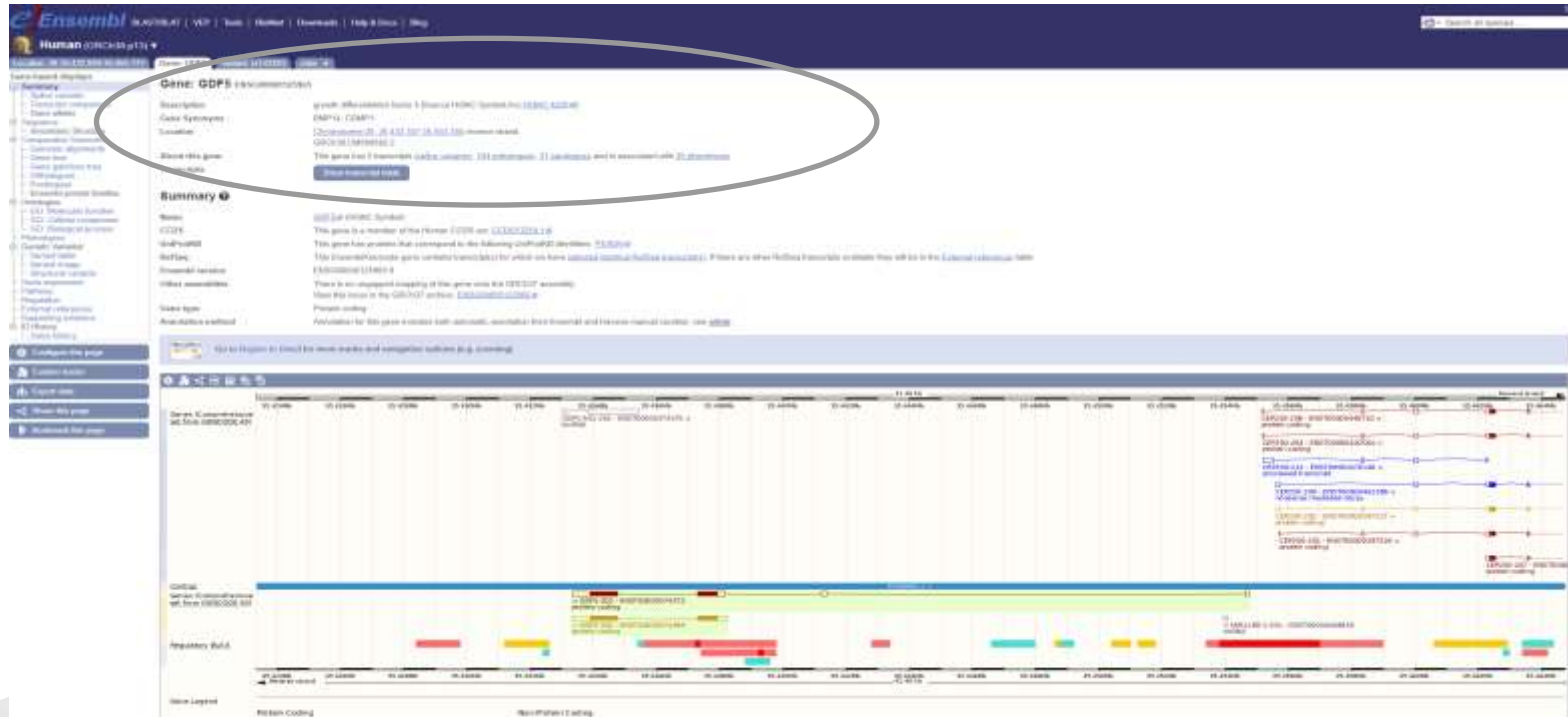
rs143383 is mentioned in the following publications:

Year	PMID	Phenotype	Title	Author(s)	Full text	Citation source
2021	33235571		Identification of key genes in osteoarthritis using bioinformatics, principal component analysis and meta-analysis.	Sun X, Guan H, Xiao L, Yao B, et al	PMC7678638	EPMC
2021	34236524		Current Evidence about Developmental Dysplasia of the Hip in Pregnancy	Simionescu AA, Cristok MM, Cristok G, Starescu ANA, et al	PMC8300860	EPMC
2021	34233342		The effect of common variants in GDF5 gene on the susceptibility to chronic postsurgical pain	Yan B, Nie H, Bu G, Yuan W, et al	PMC8247325	EPMC
2021	33937320		The Mechanisms and Functions of GDF-5 in Intervertebral Disc Degeneration	Qiao B, Cui L, Xiao C, Wang C, et al	PMC8128940	EPMC
2021	33522951		Growth differentiation factor 5 in cartilage and osteoarthritis: A possible therapeutic candidate	Sun H, Gao J, Yao X, Guo Z, et al	PMC7341218	EPMC
2021	34233380		Genetic Study of IL6, GDF5 and MMP13 in Association with Developmental Dysplasia of the Hip	Harsanyi S, Zamborsky R, Knapova L, Kolarov M, et al	PMC8300839	EPMC
2021	33973061		A Case-Control Study of Major genetic pre-disposition risk alleles in developing ODD in the Northwest US population: Effects of Gene-Gene Interactions	Volkovskiy VI, Zhu BH, Hida T, Laccoss R, et al		EPMC
2021	33963100		Frequency of Growth Differentiation Factor 3 rs143383 and angiotensin II-receptor polymorphisms in patients with hand and knee osteoarthritis in Hubeian province, Iran	Moghimi N, Nasiri S, Ghahouri F, Jafari A		EPMC
2020	31321126		A Meta-analysis Assessing the Association Between COL11A1 and GDF5 Genetic Variants and Intervertebral Disc Degeneration Susceptibility	Wu F, Huang X, Zhang Z, Shao Z		dbSNP
2020	32103174		A novel variant near LRP1P5 is associated with knee osteoarthritis in the Chinese population	Li Y, Lai F, Xia X, Zhang H, et al		EPMC,dbSNP
2020	31133748		Regulation of Gdf5 expression in joint remodelling, repair and osteoarthritis	Korea K, Colella F, Riemann AHK, Wang H, et al	PMC7026708	EPMC
2020	32523100		Association between growth differentiation factor 5 rs143383 genetic polymorphism and the risk of knee osteoarthritis among Caucasians but not Asian: a meta-analysis	Peng L, Jin S, Li J, Ouyang C, et al	PMC7308800	EPMC
2020	32381872		Developmental Dysplasia of the Hip: A Review of Etiopathogenesis, Risk Factors, and Genetic Aspects	Harsanyi S, Zamborsky R, Knapova L, Kolarov M, et al	PMC7238832	EPMC
2020	32987100		Can Implementation of Genetics and Pharmacogenomics Improve Treatment of Chronic Low Back Pain?	Burns V, Jovanovic T, Kozlovic E, Castelli KD, et al	PMC7308488	EPMC
2020	32184303		Generation and characterization of human induced pluripotent stem cells (iPSCs) from hand osteoarthritis patient derived fibroblasts	Carro-Villaverde R, Sampedro-Rodríguez C, Pavón-Ramírez A	PMC7000211	EPMC



Ensembl ... Exploring genes

➤ Let's have a look at *GDF5* (growth differentiation factor 5).



Ensembl ... Exploring variants ... Exercise 2

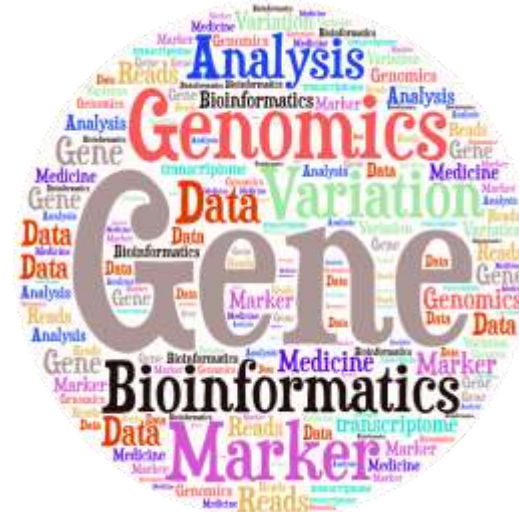
Explore rs2075650 (build 37).

1. What type of variants is?
2. Where is this variant located (chr - position - strand)?
3. Are there big differences in allele frequencies between populations?
4. What is the ancestral allele frequency in the Japanese in Tokyo (JPT) population from the 1000G set?
5. What is the least frequent genotype for this variant in the KHV population from the 1000G set?
6. In how many transcripts is this variant found?
7. Are all transcripts from the same gene?
8. How many and which consequence type are predicted for this variant?
9. How many phenotypes are associated with this variant?
10. How many phenotypes are associated with the gene that this variant is located in?
11. How many publications refer to rs2075650?



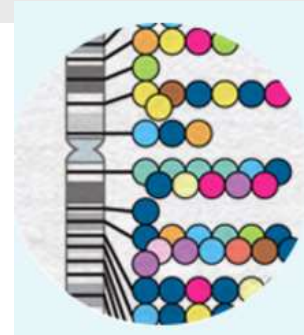
Publicly available bioinformatics resources

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



Human Traits & Diseases Databases

- GWAS catalog (<https://www.ebi.ac.uk/gwas/>)
- OMIM (<http://omim.org/>)
- Orphanet (<http://www.orpha.net>)



Human Traits & Diseases Databases

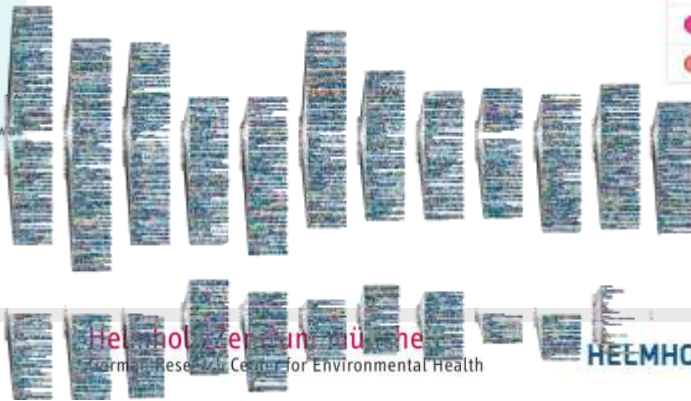
GWAS catalog

- “GWAS catalog” is a catalog of published/unpublished genome-wide association studies.



2022

Show SNPs for	
Digestive system disease	940
Cardiovascular disease	1887
Metabolic disease	1896
Immune system disease	2388
Nervous system disease	3428
Liver enzyme measurement	4383
Lipid or lipoprotein measurement	4773
Inflammatory marker measurement	485
Hematological measurement	9588
Body weights and measures	3884
Cardiovascular measurement	1876
Other measurement	37444
Response to drug	440
Biological process	2523
Cancer	3713
Other disease	3806
Other trait	2622



Human Traits & Diseases Databases

GWAS catalog



- Let's have a look at a specific variant, rs143383.

Search results for *rs143383*

V rs143383

Location: 20:35438203 **Cytogenetic region:** 20q11.22 **Most severe consequence:** 5 prime utr variant **Mapped gene(s):** GDF5

Associations **6** Studies **6**

G GDF5

Description: growth differentiation factor 5

Location: 20:35433347-35454746 **Cytogenetic region:** 20q11.22 **Biotype:** protein coding

Associations **92** Studies **67**

Human Traits & Diseases Databases

GWAS catalog

V rs143383

Location: 20:35438203 **Cytogenetic region:** 20q11.22 **Most severe consequence:** 5 prime utr variant **Mapped gene(s):** GDF5

Associations **10** Studies **10**

Available data: Associations **10** Studies **10** Traits **3** Linkage disequilibrium (LD) Download Associations

Associations **10**

Show 10 entries

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Background trait(s)	Study accession	Location
rs143383-G	2×10^{-16}	-	0.37	-	6.25 ± score increase	-	GDF5	Vertex-wise cortical surface area	cortical surface area measurement	-	GCST00095130	20:35438203
rs143383-T	5×10^{-7}	-	MR	1.049905	-	[1.03-1.07]	GDF5	Knee osteoarthritis	osteoarthritis, knee	-	GCST007000	20:35438203
rs143383-G	1×10^{-17}	-	0.3908	-	-	-	GDF5	Cortical surface area	cortical surface area measurement	-	GCST00091000	20:35438203
rs143383-T	5×10^{-18}	-	0.64	1.06287	-	[NR]	GDF5	Knee osteoarthritis	osteoarthritis, knee	-	GCST006925	20:35438203
rs143383-T	9×10^{-29}	-	MR	-	0.09367727 unit increase	[0.057-0.07]	GDF5	Height	body height	-	GCST000830	20:35438203

Showing 1 to 5 of 10 entries

Human Traits & Diseases Databases

GWAS catalog

GDF5

Description: growth differentiation factor 5

Location: 20:35433347-35454746 **Cytogenetic region:** 20q11.22 **Biotype:** protein coding

Associations **131** Studies **98**



Human Traits & Diseases Databases ...

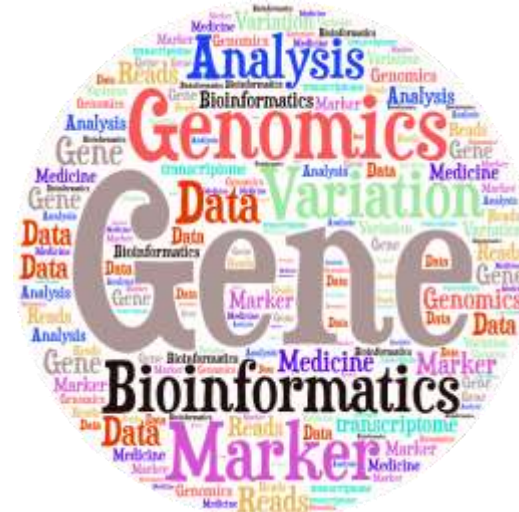
Exercise 3

1. Try to find more information about rs2075650 and the gene that is located in GWAS catalogue and OMIM databases.



Publicly available bioinformatics resources

- Human Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- **Databases of other Organisms**
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



-
- The screenshot displays the Ensembl genome browser interface for the Gdf5 gene. The top navigation bar includes the Ensembl logo and links for PLAS1501, VEP, Tools, Format, Download, Help & Docs, and Blog. The main header shows the gene name 'Gdf5' and its chromosome location 'chr10:100,000,000-100,000,000'. The left sidebar contains a 'Summary' section with links to various gene features and a 'Genomic track' section with links to various genomic tracks. The main content area is divided into two columns. The left column contains a table with columns for Name, Transcript ID, bp, Protein, Biotype, CDS, and RefSeq. The right column contains a 'Summary' section with a description of the gene, its location, and its function. The bottom section of the page features a genomic track visualization showing the gene structure, including exons and introns, and a protein structure visualization showing the protein domains and structure.

Databases of other Organisms

- Mouse: <http://www.informatics.jax.org/>

The screenshot shows the Mouse Genome Informatics (MGi) website. The header includes the MGI logo, the text "Mouse Genome Informatics", and navigation links: Search, Download, More Resources, Submit Data, Find Mice (IMSR), Analysis Tools, Contact Us, and Browser. A sidebar on the left lists various search and analysis tools: Genes, Phenotypes & Mutant Alleles, Human-Mouse Disease Connection, Gene Expression Database (GXD), Recombinase (cre), Function, Strains, SNPs & Polymorphisms, Vertebrate Homology, Mouse Models of Human Cancer, Batch Data and Analysis Tools, and Nomenclature. A "Getting Started" section provides links to an introduction to mouse genetics, how to use MGI (text & video), and a Cre Portal Tutorial. The main content area features a "Quick Search" box, a "NEW: MOUSE RESOURCES FOR COVID-19 RESEARCH" banner, a description of MGI as an international database resource, and a large graphic with the text "HTTPS MGI is now supporting the more secure HTTPS protocol". Below this is a "What's new at MGI" section with a list of updates dated September 11, 2021, including human-mouse disease connection search refinements, improved representation of the pseudautosomal region, QTL Detail Pages, and updates from the Global Biobase Coalition (GBC) and the Alliance of Genome Resources.

MGi About Help FAQ

Mouse Genome Informatics

Search Download More Resources Submit Data Find Mice (IMSR) Analysis Tools Contact Us Browser

Keywords, Symbols, or IDs (exact phrase) Quick Search

Type your search here

Or use topic specific search and analysis tools:

- Genes
- Phenotypes & Mutant Alleles
- Human-Mouse Disease Connection
- Gene Expression Database (GXD)
- Recombinase (cre)
- Function
- Strains, SNPs & Polymorphisms
- Vertebrate Homology
- Mouse Models of Human Cancer
- Batch Data and Analysis Tools
- Nomenclature

Getting Started:

- Introduction to mouse genetics
- How to use MGI (Text & Video)
- Cre Portal Tutorial

<https://www.informatics.jax.org/humanDisease.shtml>

****NEW: MOUSE RESOURCES FOR COVID-19 RESEARCH****

MGi is the international database resource for the laboratory mouse, providing integrated genetic, genomic, and biological data to facilitate the study of human health and disease.

[About Us](#) [MGi Publications](#) [Databases](#)

HTTPS

MGi is now supporting the more secure HTTPS protocol

What's new at MGI updated September 11, 2021

- Human - Mouse: Disease Connection search refinements. [Read more...](#)
- Improved representation of the pseudautosomal region. [Read more...](#)
- QTL Detail Pages now include candidate genes and QTL interactions. [Read more...](#)
- The Global Biobase Coalition (GBC) names MGI a Global Core Biobase Resource. [Read more...](#)
- A new tool allows searching for genes by their expression profile. [Read more...](#)
- Human gene to disease annotations are now from the Alliance of Genome Resources. [Read more...](#)

Databases of other Organisms

- Mouse: <https://www.mousephenotype.org/>



ABOUT THE IMPC

DATA

HUMAN DISEASES

PUBLICATIONS

NEWS

BLOG

**We are creating a comprehensive catalogue of
mammalian gene function**

GENES

PHENOTYPES

HELP, NEWS, BLOG

Search All 8707 Knockout Data...



What you need to know about IMPC data

Databases of other Organisms

- Zebrafish: <http://zfin.org/>

ZFIN Research Genomics Resources Community Support Sign in

Search expert curated zebrafish data

Any heart contraction abnormal Search

Genes
Search for genes, transcripts, clones, and other markers

Expression
Search for gene expression data, and annotated images

About ZFIN
The Zebrafish Information Network (ZFIN) is the database of genetic and genomic data for the zebrafish (*Danio rerio*) as a model organism. ZFIN provides a wide array of expertly curated, organized and cross-referenced zebrafish research data.
[Learn More](#)

Mutants/Tg
Search for mutants, knockdowns, transgenics, and affected phenotypes

Antibodies
Search for antibodies by gene, labeled anatomy, and other attributes

BLAST
Align nucleotide and protein sequences with zebrafish datasets

Publications
Search for zebrafish research publications and scientific literature

Additional Resources
[Data Mining](#)
[ZebrafishMine](#) [BioMart](#)

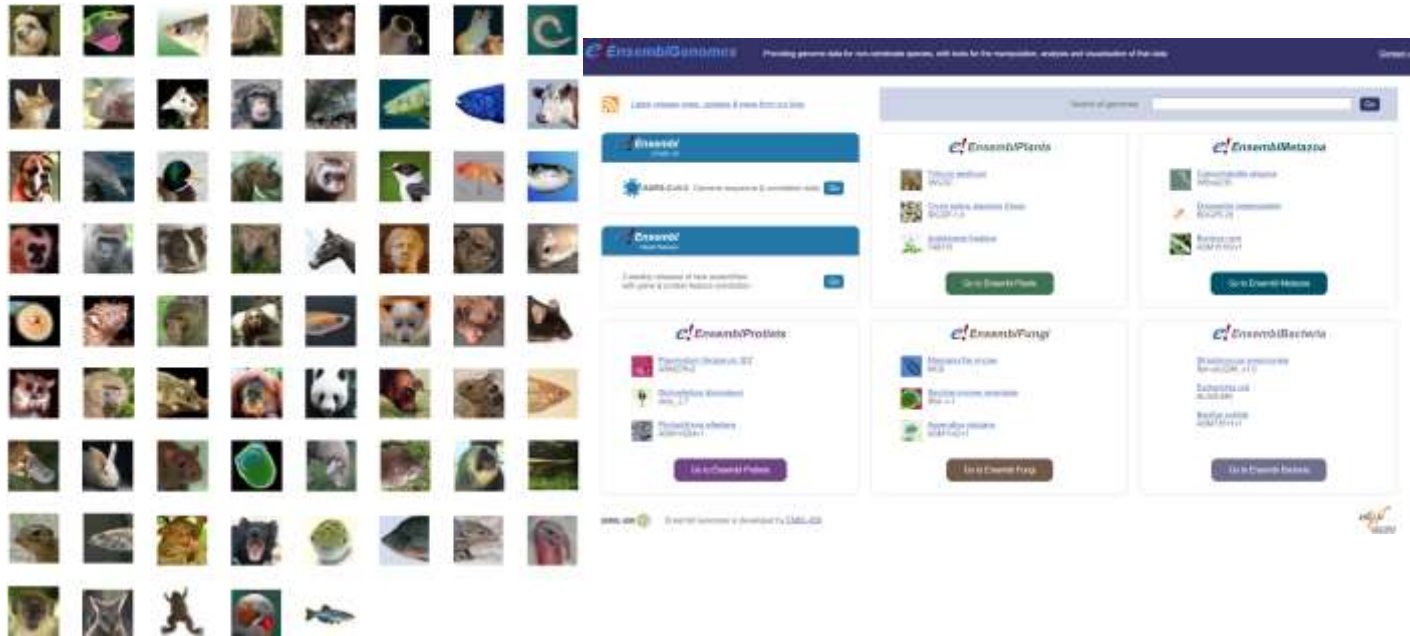
Databases of other Organisms

- Drosophila: <http://flybase.org/>

The screenshot shows the FlyBase website, a database for Drosophila genes and genomes. The header includes the FlyBase logo, the tagline "A Database of Drosophila Genes & Genomes", and a navigation bar with links: Home, Tools, Downloads, Lyrics, Community, Species, About, Help, Archives. A search bar on the right contains the text "229" and a "Jump to Gene" button. Below the navigation bar is a row of icons for various tools and resources: Blast, GBrowse, JBrowse, CRISPR, RNA-Seq, GO Phenotype Anatomy Disease, ImageBrowse, and Batch Download. The main content area is divided into two sections. On the left, there is a "Using JBrowse on FlyBase" video player and a sidebar with links to "FAST-TRACK YOUR PAPER", "FLYBASE NEWS", "FLY BOARD", "COMMUNITY NEWS", "MEETINGS COURSES", and "FLYBOOK". On the right, there is a "QuickSearch" section with tabs for Human Disease, Protein Domains, Gene Groups, Pathways, GO, and Data Class. Below these tabs is a search bar with the text "Search FlyBase" and a "Search" button. The bottom section of the page features a grid of ten icons representing different tissues and organs: Adipose, Circulatory, Excretory, Muscle, Imaginal Precursor, Integumentary, Tracheal, Digestive, Nervous, and Reproductive. Each icon is accompanied by its respective label.

Databases of other Organisms

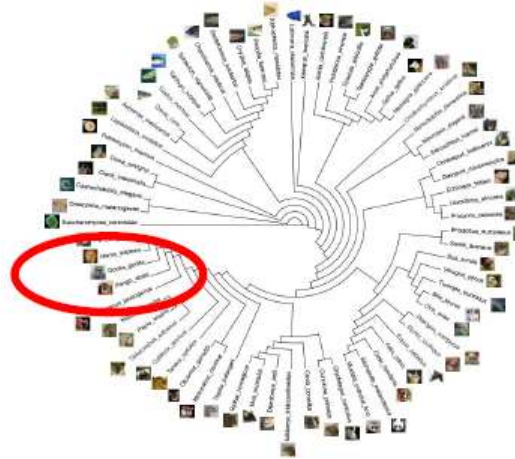
- Ensembl : >300 vertebrate species (<https://www.ensembl.org/info/about/species.html>) & >45000 genomes from non-vertebrate species (<http://ensemblgenomes.org/>)



Databases of other Organisms ...

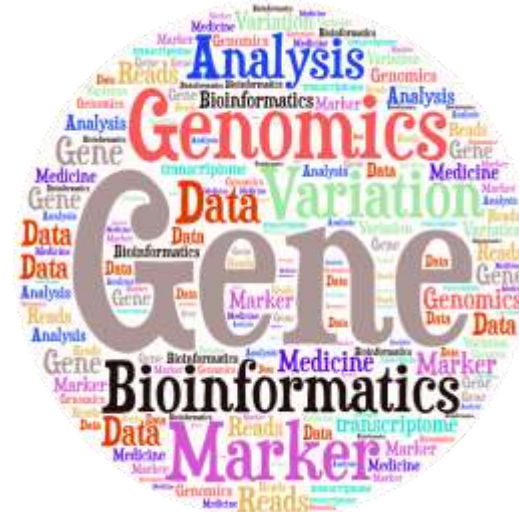
Exercise 4

- Explore rs2075650 in the databases of other organisms.
 - *Hint: you can search by rsID or (mapped) gene id or phenotype.*



Publicly available bioinformatics resources

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- **Gene Expression Databases**
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



Gene Expression Databases

- Genotype-Tissue Expression (GTEx) : Correlations between genotype and tissue-specific gene expression (<https://www.gtexportal.org/home/>).

The screenshot shows the GTEx Portal homepage. At the top, there's a navigation bar with links like Home, Downloads, Expression, Single Cell, GTL, IGV Browser, Tissues & Histology, and Documentation. A search bar is also present. Below the navigation bar, there's a banner for "miRNA-Seq Data released". The main content area is divided into two columns: "Resource Overview" and "Explore GTEx".

Resource Overview

Current Release (V8)

Tissue & Sample Statistics
Tissue Sampling Info (Anatomogram)
Access & Download Data
Release History
How to cite GTEx?

The Genotype-Tissue Expression (GTEx) project is an ongoing effort to build a comprehensive public resource to study tissue-specific gene expression and regulation. Samples were collected from 58 non-diseased tissue sites across nearly 1000 individuals, primarily for molecular studies including WGS, WES, and RNA-Seq. Human samples are available from the GTEx Biobank. The GTEx Portal provides open access to data including gene expression, GTLs, and histology images.

Developmental GTEx

The Developmental Genotype-Tissue Expression (dGTEx) Project is a new effort to study developmentally specific genetic effects on gene expression. The major goal of the project is to establish a molecular and data analysis resource as well as a tissue bank to study the regulation of gene expression in multiple relatively healthy reference neonates, pediatrics, and adolescent tissues, building on the Genotype-Tissue Expression (GTEx) project.

Explore GTEx

Browse

By gene ID: Browse and search all data by gene
By variant or rs ID: Browse and search all data by variant
By Tissue: Browse and search all data by tissue
Histology Viewer: Browse and search GTEx histology images

Single Cell

Data Overview: Learn more about available single cell data
Multi-Gene Single Cell Query: Browse and search single cell expression by gene and tissue

Expression

Multi-Gene Query: Browse and search expression by gene and tissue
Transcript Browser: Visualize transcript expression and isoform structures

GTL

Locus Browser (Gene-centric): Visualize GTLs by gene in the Locus Browser
Locus Browser (Variant-centric): Visualize GTLs by variant in the Locus Browser VC (Variant-centric)

Gene Expression Databases

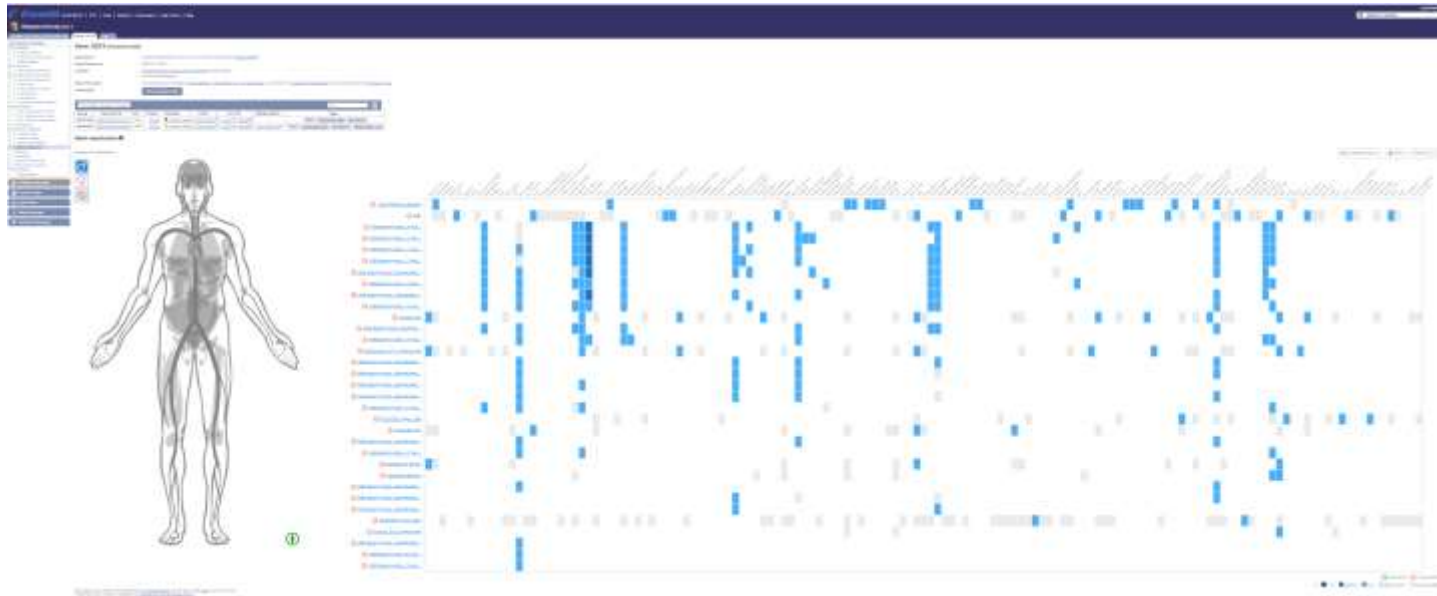
- Expression Atlas: Differential and Baseline Expression (<http://www.ebi.ac.uk/gxa/home> or through Ensembl).

The screenshot shows the Expression Atlas website. The header includes the EMBL-EBI logo and navigation links. The main banner features the 'Expression Atlas' title and a search bar. Below the banner, a navigation bar lists various links. The main content area displays a search bar with a 'Search' button and a 'Clear' button. Below the search bar, there are tabs for 'Animals', 'Plants', and 'Fungi'. A grid of species cards is shown, each with an icon, the species name, and the number of experiments, baseline, and differential expression studies.

Species	Experiments	Baseline	Differential
Homo sapiens	1448	99	1390
Mus musculus	1153	46	1107
Rattus norvegicus	152	3	149
Drosophila melanogaster	140	4	136
Gallus gallus	36	3	33
Caenorhabditis elegans	29	1	28

Gene Expression Databases

- Expression Atlas: Differential and Baseline Expression (<http://www.ebi.ac.uk/gxa/home> or through Ensembl).



Gene Expression Databases ...

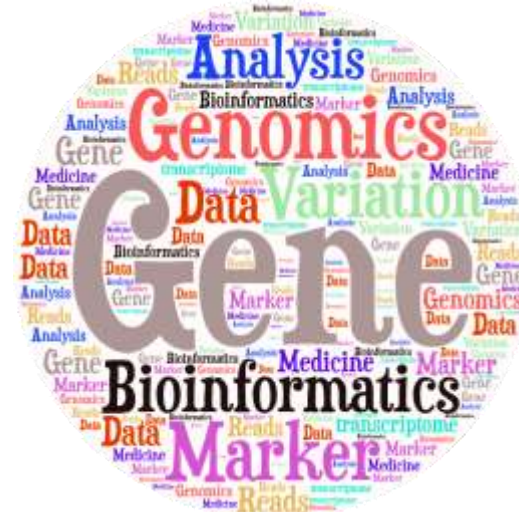
Exercise 5

- Does rs2075650 affect the expression of the gene?
 - Hint: Use GTEx portal
- In which tissue *TOMM40* is expressed? Any connections with Alzheimer disease?
 - Hint: Use Expression Atlas



Publications Database

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



- PubMed (<https://www.ncbi.nlm.nih.gov/pubmed>) comprises over 34 million citations.
- How do I search PubMed?
 - Identify the key concepts for your search.
 - Enter the terms* in the search box.
 - Keywords
 - Author name
 - Journal name
 -
 - Suggestions will display as you type your search terms.
 - Click Search.



Publications Database

- Anatomy of the summary results:

- ☐ [Policy Issues in the Development and Adoption of Biomarkers for Molecularly Targeted Cancer Therapies: Workshop Summary.](#)
1. National Cancer Policy Forum, Board on Health Care Services, Institute of Medicine.
Washington (DC): National Academies Press (US); 2015.
PMID: 25855848 [Free Books & Documents](#)
[Similar articles](#)
- ☐ [Four-wave mixing experiments with extreme ultraviolet transient gratings.](#)
2. Bencivenga F, Cucini R, Capotondi F, Battistoni A, Mincigrucci R, Giangrisostomi E, Gessini A, Manfreda M, Nikolov IP, Pedersoli E, Principi E, Svetina C, Parisse P, Casolari F, Danailov MB, Kiskinova M, Masciovecchio C.
Nature. 2015 Apr 9;520(7546):205-8. doi: 10.1038/nature14341.
PMID: 25855456
[Similar articles](#)
- ☐ [Molecular imaging of angiogenesis after myocardial infarction by \(111\)In-DTPA-cNGR and \(99m\)Tc-sestamibi dual-isotope myocardial SPECT.](#)
3. Hendrikx G, De Saint-Hubert M, Dijkgraaf I, Bauwens M, Douma K, Wiertz R, Pooters I, Van den Akker NM, Hackeng TM, Post MJ, Mottaghy FM.
EJNMMI Res. 2015 Jan 28;5:2. doi: 10.1186/s13550-015-0081-7. eCollection 2015.
PMID: 25853008 [Free PMC Article](#)
[Similar articles](#)

title

authors

journal title
abbreviation

volume & issue

e-pagination

publication date



Publications Database ... Exercise 6

- Try to find more information about rs2075650 and the gene that is located.
- Can you find a paper for which:
 - “Lopes” is one of the authors
 - Dealing with “osteoarthritis”
 - The name of the journal is “Lancet”



Variant:

Ensembl:

- ✓ position,
- ✓ consequences (transcript and regulatory region)
- ✓ 1kG frequencies,
- ✓ GWAS signals around the variant,
- ✓ regulatory marks around the variant
- ✓ nearby genes.

ExAC: allele frequencies

GTEX: associated gene.

Pubmed: papers where the variant is cited.

Gene:

Ensembl:

- ✓ Position
- ✓ Associated phenotypes
- ✓ Gene ontology annotations

Uniprot:

- ✓ Function, Tissue, Localization, Disease

OMIM:

- ✓ gene annotation
- ✓ associated diseases

GWAS catalog:

- ✓ Associated variants and association details.

Gene Expression Atlas:

- ✓ Tissues in which expression of the gene was observed

GTEX:

- ✓ List of variants that affects the expression of the gene.

Mouse Genomics Informatics:

- ✓ Mouse phenotype of the gene.



Dr Daniel Suveges
Senior Bioinformatician

Team144 pipeline

Variant details

General information:

rsID: [rs143384](#)
Chromosome:start-end: [20:34015756-34015756](#)
Allele string: A/G
MAF: 0.438898
Consequence: 5_prime_UTR_variant
Variation type: SNP
gerp (average gerp): 5.04 (3.497)
GWAVA score: 0.79
Ancestral allele: G
Minor allele: A
Synonyms: rs431839, rs61433024, rs765106, rs3748435, rs17422934
Phenotypes (risk allele): [Height\(A\)](#),
[Infant length\(G\)](#),
[Height\(G\)](#)

GWAS signals

GWAS signals within 1Mbp of the variant:

rsID	SNPID	Distance	Trait	p-value	PMID
rs17330467	chr20:33545616	-481kbp	Hemostatic factors and hematological phenotypes	4E-34	22443381
rs11906180	chr20:33565755	-461kbp	Anticoagulant levels	1E-6	22216198
rs1335166	chr20:33718706	-308kbp	Height	1.8E-29	25282101
rs2205888	chr20:33722863	-303kbp	Prothrombin time	5E-13	22703881
rs6120849	chr20:33730387	-296kbp	Protein C levels	7E-37	20802025
rs6988725	chr20:33745676	-281kbp	Hemostatic factors and hematological phenotypes	4E-34	22443381
rs6900278	chr20:33753262	-273kbp	Hemostatic factors and hematological phenotypes	4E-34	22443381
rs867186	chr20:33764554	-262kbp	Anticoagulant levels	4E-9	22216198
rs867186	chr20:33764554	-262kbp	Cosignation factor levels	6E-37	20231535
rs867186	chr20:33764554	-262kbp	D-dimer levels	4E-6	21502571
rs867186	chr20:33764554	-262kbp	Hemostatic factors and hematological phenotypes	2E-6	22443381
rs867186	chr20:33764554	-262kbp	Protein C levels	1E-64	24376901
rs11167260	chr20:33775200	-251kbp	Amniotrophic lateral sclerosis	4E-6	24529757



Dr Daniel Suveges
Senior Bioinformatician

HelmholtzZentrum münchen

German Research Center for Environmental Health

Thank you!

HELMHOLTZ RESEARCH FOR
GRAND CHALLENGES

What is a genome assembly?

Sequence reads

```
CGGCCTTTGGGCTCCGCCTTCAGCTCAAGA
CAGCTGTCCCAGATGAC    ACTTAACTTCCCTCCCAGCTGTCC
GGGCTCCGCCTTCAGCTC   AACTTCCCTCCCAGCT    TCCCAGCTGTCCCAGATGACGCCATC
CGGCCTTTGGGCTCC      CAGATGACGCC    TCCGCCTTCAGCTCAAGACTTAACTTC
```

Match up overlaps

```
CGGCCTTTGGGCTCCGCCTTCAGCTCAAGA  AACTTCCCTCCCAGCT    CAGATGACGCC
                                TCCGCCTTCAGCTCAAGACTTAACTTC  TCCCAGCTGTCCCAGATGACGCCAT
                                GGGCTCCGCCTTCAGCTC    ACTTAACTTCCCTCCCAGCTGTCC
CGGCCTTTGGGCTCC                                CAGCTGTCCCAGATGAC
```

Genome assembly

```
CGGCCTTTGGGCTCCGCCTTCAGCTCAAGACTTAACTTCCCTCCCAGCTGTCCCAGATGACGCCAT
```

Ensembl ... the front page ... Exercise 1



1. Go to the species homepage for **Giant Panda**. What is the name of the genome assembly for Panda?
 - ASM200744v2, INSDC Assembly [GCA_002007445.2](#), Apr 2020
2. How long is the Panda genome (in bp)? How many coding genes have been annotated?
 - 2,444,060,653 & 20,857
3. Repeat for **Human**.
 - GRCh38.p13 (Genome Reference Consortium Human Build 38), INSDC Assembly GCA_000001405.28, Dec 2013
 - 3,096,649,726
 - 20,471 (incl 662 readthrough)



Ensembl ... Exploring variants ... Exercise 2

Explore rs2075650 (build 37).

1. What type of variants is? **SNP**
2. Where is this variant located (chr - position - strand)? **Chromosome 19:44892362 (forward strand)**
3. Are there big differences in allele frequencies between populations? **Not really. Max 8% in some sub-African pop and 10% in som AMR sub-pop too.**
4. What is the ancestral allele frequency in the Japanese in Tokyo (JPT) population from the 1000G set? **12%**
5. What is the least frequent genotype for this variant in the KHV population from the 1000G set? **G|G**
6. In how many transcripts is this variant found? **5**
7. Are all transcripts from the same gene? Yes, **TOMM40**
8. How many and which consequence type are predicted for this variant? **1, Intron**
9. How many phenotypes are associated with this variant? **67**
10. How many phenotypes are associated with the gene that this variant is located in? **1**
11. How many publications refer to rs2075650? **313**